

EV Challenge Team Report

1. Team Information

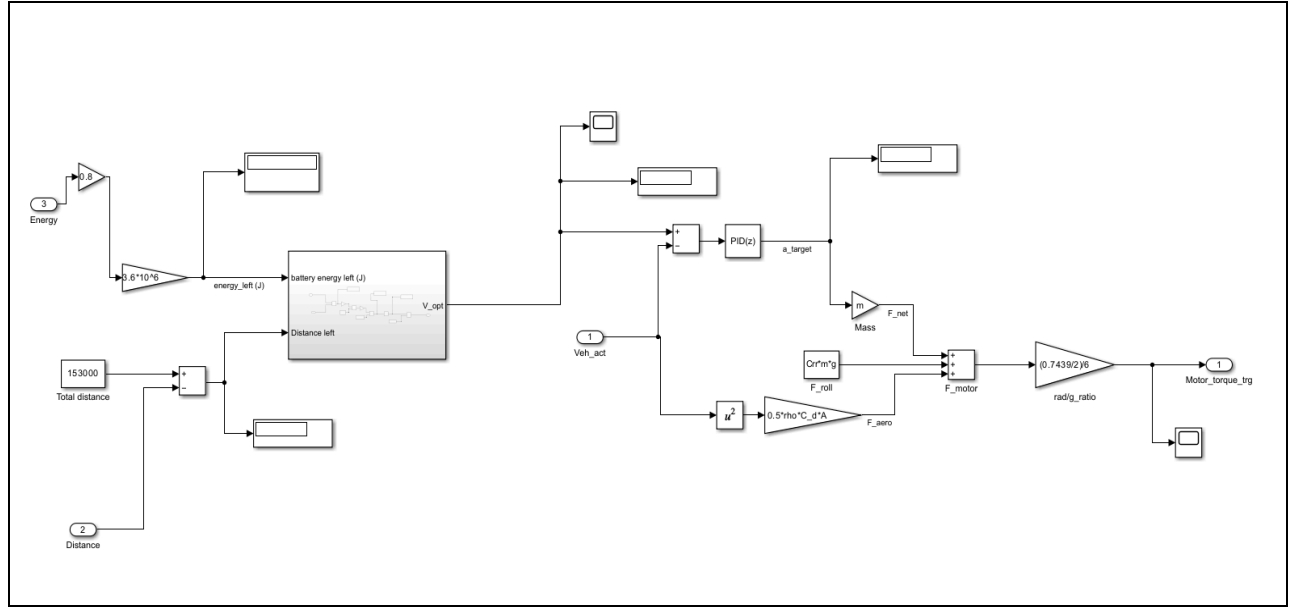
- **Team Name:** IITGRacing
- **Team Members:**
 - Ronak
 - Ryan Sequeira
 - Bedabrat Barma
 - K Aravindhana
 - Aayush Kumar
- **Mentor Name & Affiliation:** Mohammed Mosaddique Nawaz, IIT Guwahati EEE Department Professor

2. FMU Details

The FMU was generated using MATLAB Simulink's FMI Export Toolbox, compliant with FMI 2.0 for Co-Simulation.

Parameters Used in FMU:

Parameter	Symbol	Value	Units
Air density	ρ	1.225	kg/m ³
Drivetrain Efficiency	η	0.9	-
Gravity	g	9.81	m/s ²
Frontal Area	A	2.48	m ²
Vehicle Mass	m	1400	kg
Rolling Resistance	C_{rr}	0.01	-
Drag Coefficient	C_d	0.31	-
Maximum Velocity	v_{max_veh}	16.667	m/s
Gear Ratio	G	6	-



Optimal Speed Model:

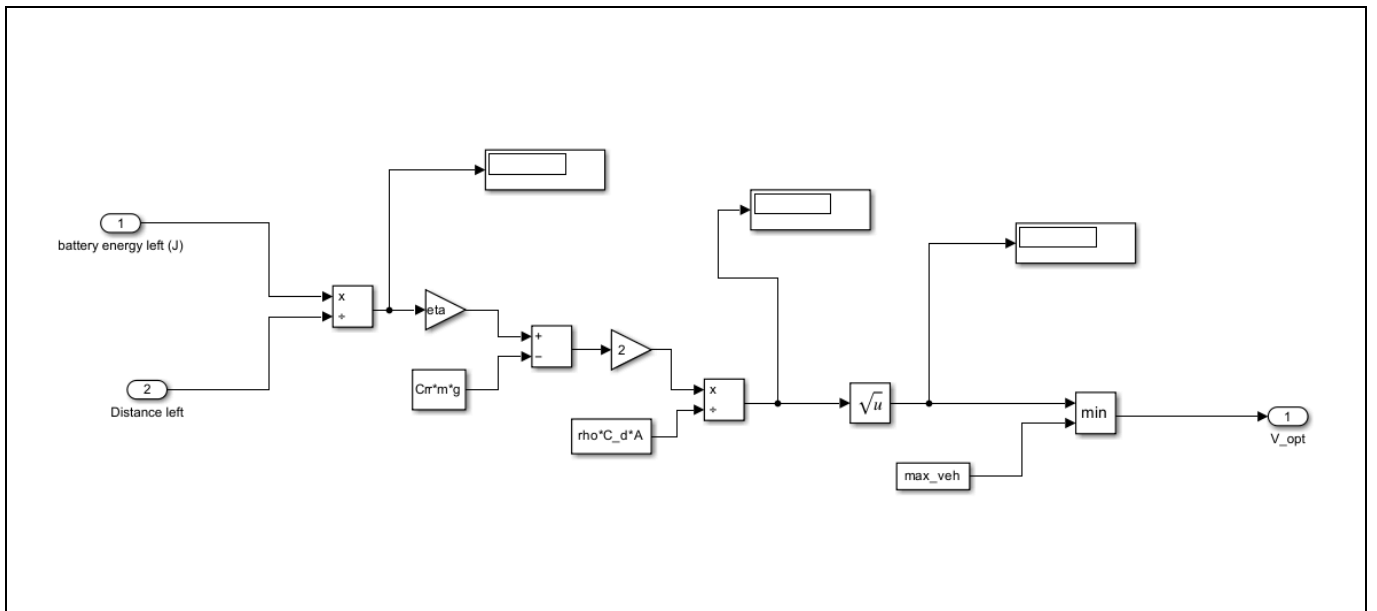
The Simulink model computes **V_{opt}** based on:

$$V_{opt} = \sqrt{\{[(E/d) \cdot \eta - C_{rr} \cdot m \cdot g] / (0.5 \cdot \rho \cdot C_d \cdot A)\}} \quad - (1)$$

*The **V_{opt}** value is constrained to not exceed the value of **max_veh** (16.667 m/s).*

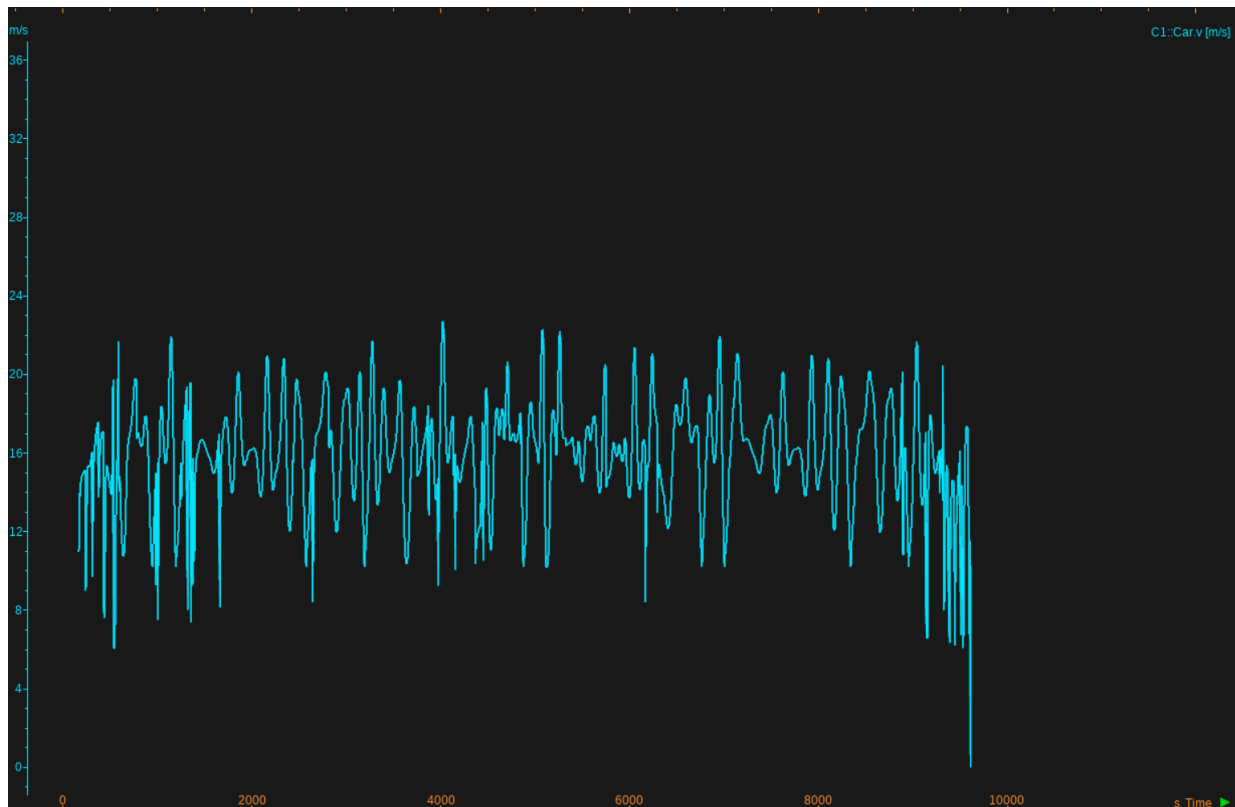
The Simulink model to compute the **Motor Force** based on:

$$F_{motor} = C_{rr} * m * g + 1/2 * \rho * C_d * A * veh_{actual}^2 + m * a \quad - (2)$$



The total distance of **153 km** was covered in **9612.2 seconds (2.67 hours)**, with **19% SoC** remaining in the end.

Velocity v/s Time:



Phase 2:

1. System Architecture

1.1 Overview

The integrated system couples three components:

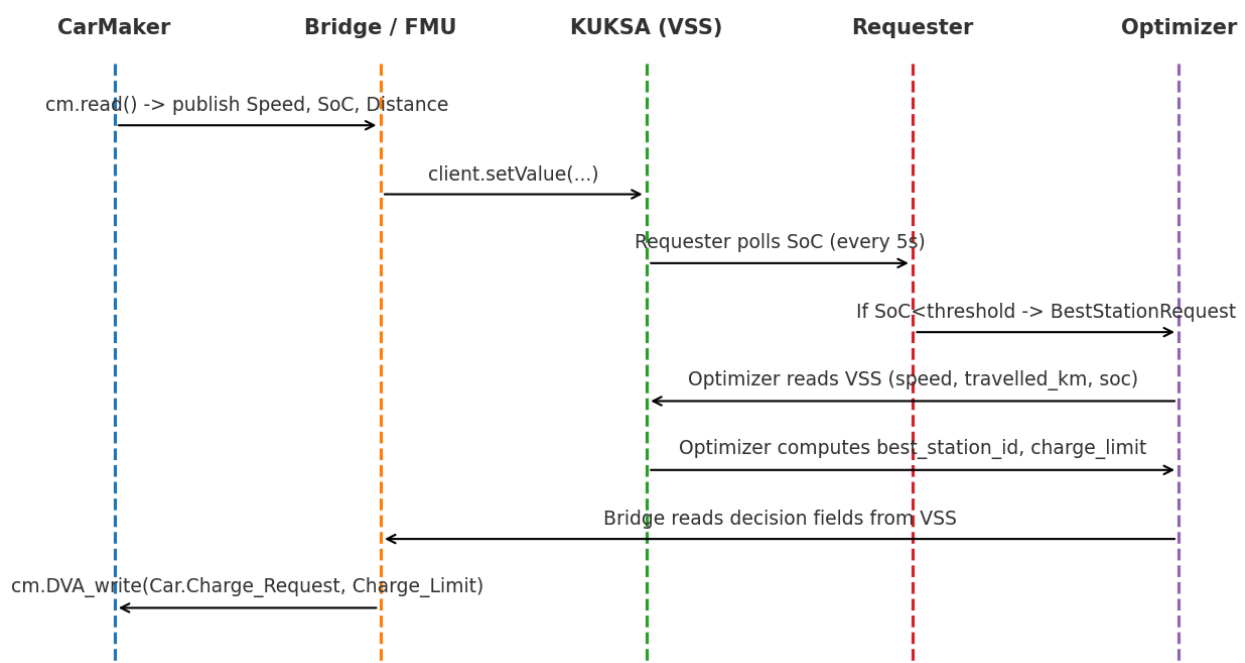
- CarMaker — high-fidelity vehicle dynamics and powertrain simulation (source of motor

torque, total force, vehicle speed, distance traveled, and battery SoC).

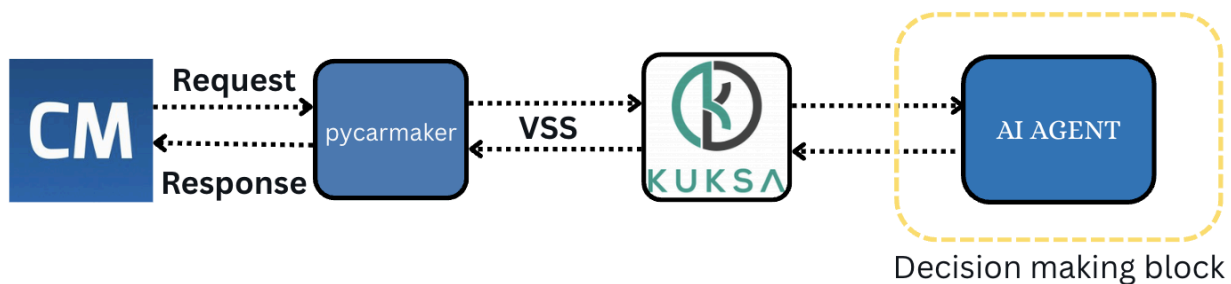
- KUKSA Databroker (VSS) — middleware for publishing and subscribing to standardized vehicle signals (Vehicle Signal Specification).
- AI Agent System — a multi-agent optimizer (Requester and Optimizer agents implemented with **uagents**) that consumes vehicle state from KUKSA, computes optimal charging decisions, and writes results back to KUKSA for CarMaker consumption.

Data flow (logical):

CarMaker → Bridge (FMU / PyCarMaker script) → KUKSA → AI Agents → KUKSA → Bridge → CarMaker



1.2 Implementation mapping (code references)



- **CarMaker → KUKSA (bridge):** implemented in the second script using `pycarmaker` and `kuksa_client`.

Key actions:

- `cm.read()` obtains CarMaker quantities: `PT.BCU.BattHV.SOC`, `Car.Distance`, `Car.v`.
 - `client.setValue('Vehicle.Speed', Vehicle_Speed.data)` etc. pushes the values into KUKSA.
- **KUKSA → AI Agents:** `AI_Agents.py` subscribes to the KUKSA VSS signals (`Vehicle.Speed`, `Vehicle.TraveledDistance`, `Vehicle.Acceleration.Longitudinal`) via the `VSSSubscription` helper and reads them into the agent logic.
- **Decision publishing:** `Car.Charge_Request` is used to generate a request (message) that activates the `optimizer` agent, which writes the selected `best_station_id` and `Car.Charge_Limit`, these values are then published to KUKSA.
- **Agent orchestration:** `Bureau()` is used to run both `optimizer` and `requester` agents concurrently.

2. AI Logic Design

2.1 Objective

Minimize total trip time (travel time + charging time) while **guaranteeing safe reachability** and a **minimum residual SoC** at the route end above a threshold of 20% SoC.

The optimizer returns: (1) the best charging station ID and

(2) the required Charge Limit (SoC).

2.2 Design

- AI Agent subscribes to VSS values.
- Requester Agent checks if the **current SoC of the vehicle** < **25%** or not.
- If True Requester Agent triggers Optimizer Agent(message).
- Optimizer Agent calculates the best station based on the 4 parameters:
 - Time required to reach the station
 - SoC at reaching station(must be greater than 15%)
 - Time required to reach the destination
 - Final SoC at destination

- Along with this, the Charge Limit is also calculated to optimize the SoC at which car is charged up to.
- Based on these parameters, the **station_id** and **Charge_Limit** are calculated and published to KUKSA via VSS.

3. Bridge Highlights

3.1 Purpose and role

The PyCarMaker bridge encapsulates the real-time data translation between CarMaker and KUKSA and guarantees temporal alignment of simulation state and AI decisions.

4. Performance Metrics and Evaluation Plan

This section lists the recommended performance metrics, their measurement method, and reasonable target values for simulation-grade evaluation.

4.1 Metrics and how to measure

1. End-to-end Latency

- Definition: time from CarMaker publishing a new vehicle state → optimizer decision is written back to CarMaker.
- Measurement: timestamp at bridge publish to KUKSA, timestamp at optimizer response write to KUKSA, and timestamp when bridge reads decision and writes to CarMaker. End-to-end = last – first.
- Target: < 200 ms for interactive simulation (adjust based on required control loop frequency).

2. Decision Correctness (Feasibility)

- Definition: fraction of physically feasible optimizer decisions (vehicle reaches chosen station without violating **MIN_SOC_ARRIVAL**).
- Measurement: simulate many drive cycles and check if **SoC_at_arrival >= MIN_SOC_ARRIVAL** and route completes with **final_soc >= MIN_SOC_FINISH**.
- Target: 100% feasibility (optimizer must filter infeasible stations).

3. SoC Estimation Error

- Definition: average absolute difference between AI-estimated SoC trajectory and CarMaker's simulated SoC.
- Measurement: Log SoC estimate used by optimizer and ground-truth SoC from CarMaker over multiple runs. Compute mean absolute error (MAE).
- Recommended target: $MAE < 2\text{--}5$ percentage points depending on consumption model fidelity. Lower if torque/force energy model is used.

4. Optimization Overhead

- Definition: CPU time or wall-clock time taken by optimizer to evaluate all candidate stations per request.
- Measurement: time per `handle_request` execution.
- Target: < 100 ms per request for typical station counts (scales with number of candidate stations).

5. Robustness / Stability

- Definition: system operation without crashes, missed messages, or deadlocks across extended runs.
- Measurement: run long drive-cycle scenarios (hours of simulated time), count restarts/errors, and measure message loss.
- Target: Zero crashes and no lost decisions under nominal load.

6. Scalability

- Definition: how performance changes with more stations, more vehicles, or more agents.
- Measurement: increase `len(stations)`, add parallel vehicle instances, measure latency and optimizer time.
- Metric: latency vs. station count plot, and CPU use vs. number of vehicles.

5. Team Experience

Phase 1 (Simulink):

- Built vehicle & powertrain model in Simulink.
- Optimized motor torque for efficiency while meeting performance targets.
- Validated energy consumption baseline for charging calculations.

Phase 2 (AI Integration):

- Connected CarMaker/Simulink signals (SoC, speed, distance) to KUKSA VSS.
- Designed AI agents (Requester + Optimizer) for charging-station selection.
- Returned optimal charging station ID & charge target to simulator in real time.

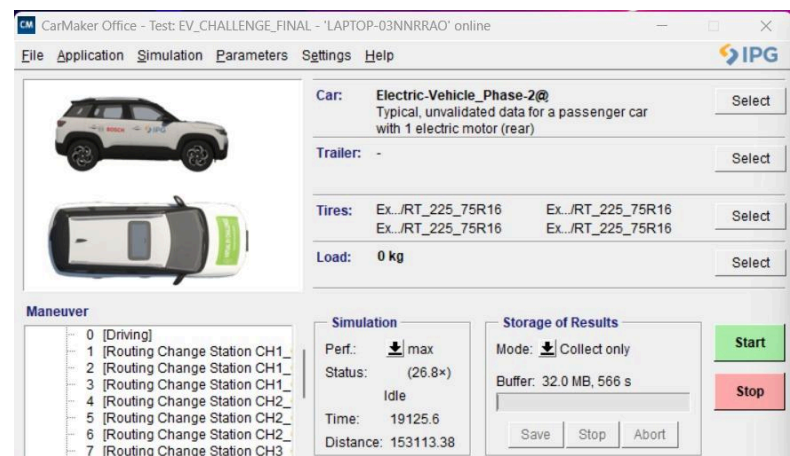
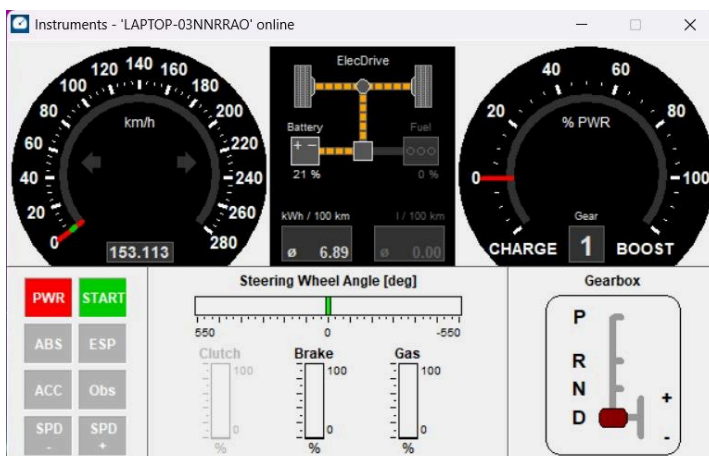
Lessons Learned:

- Torque-based SoC estimation is more reliable than distance-based.
- Event-driven FMUs improve response over polling loops.
- Clear signal mapping in VSS is critical for debugging.
- Splitting monitoring and computation improves responsiveness.

6. Results

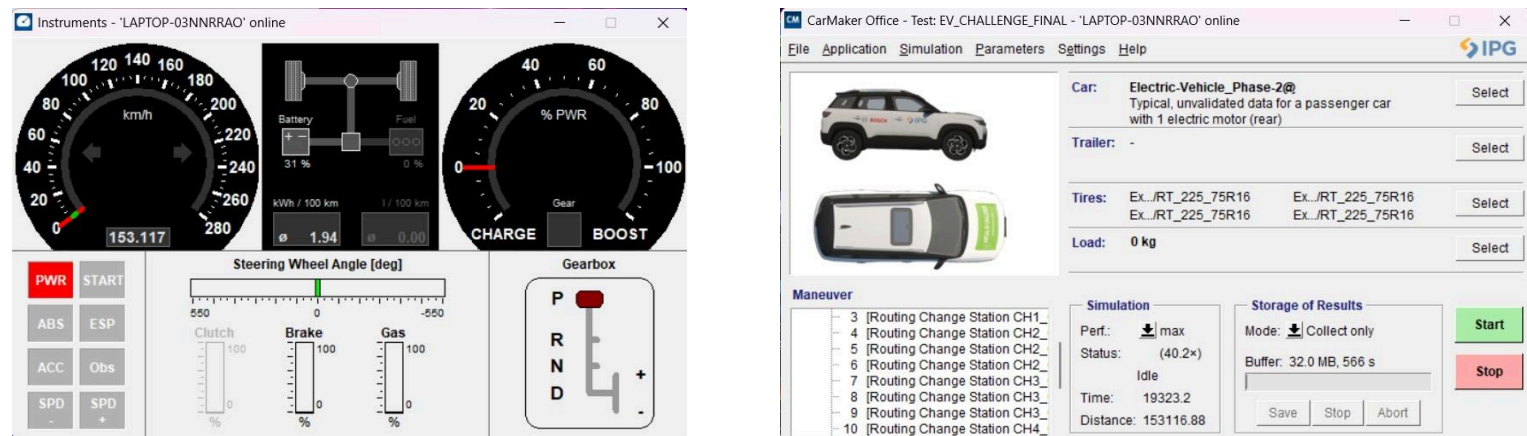
Simulation Results for 50% SoC:

```
22.22449493408203
INFO: [requester]: SoC low (22.22%), requesting optimizer
INFO: [optimizer]: Received best station request
INFO: [requester]: Booking successful → Best station 14 with charge limit 41.70875
INFO: [requester]: station_id=14 distance_to_station_km=18.82 time_to_station_h=0.29 soc_at_arrival_percent=15.17 charging_time_h=0.53 time_to_destination_h=0.89 final_soc_percent=20.0 total_trip_time_h=1.71
```



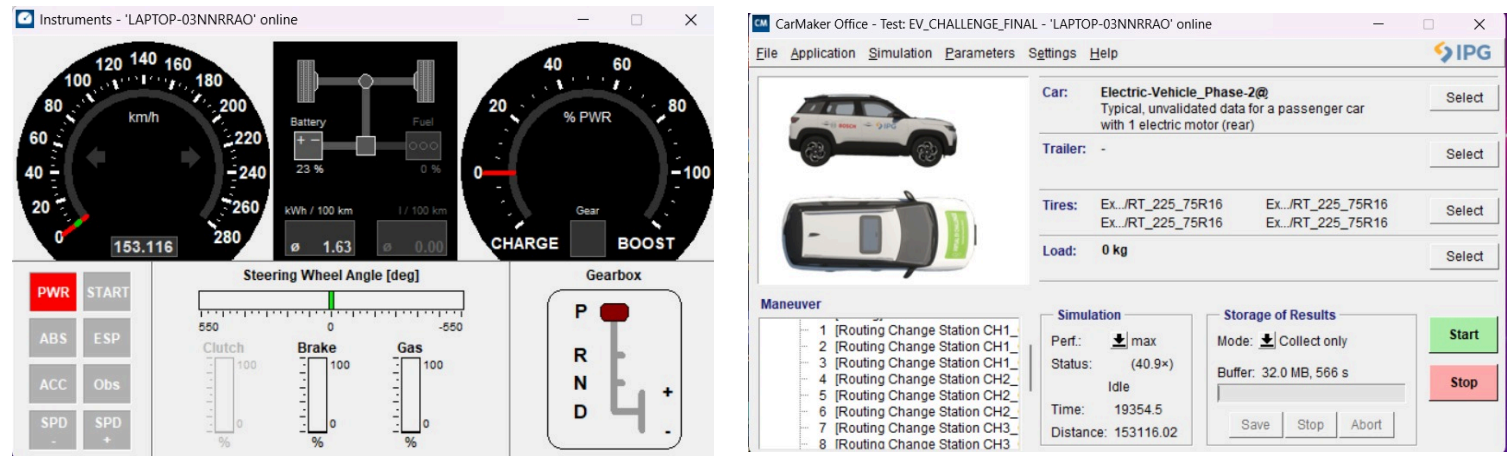
Simulation Results for 40% SoC:

```
25.285921096801758
23.55013656616211
INFO: [requester]: SoC low (23.55%), requesting optimizer
INFO: [optimizer]: Received best station request
INFO: [requester]: Booking successful → Best station 11 with charge limit 66.61125
INFO: [requester]: station_id=11 distance_to_station_km=13.6 time_to_station_h=0.21 soc_at_arrival_percent=18.45 charging_time_h=0.96 time_to_destination_h=1.5 final_soc_percent=30.0 total_trip_time_h=2.67
```



Simulation Results for 30% SoC:

```
25.317874908447266
23.93931770324707
INFO: [requester]: SoC low (23.94%), requesting optimizer
INFO: [optimizer]: Received best station request
INFO: [requester]: Booking successful → Best station 8 with charge limit 68.26249999999999
INFO: [requester]: station_id=8 distance_to_station_km=10.35 time_to_station_h=0.16 soc_at_arrival_percent=20.06 charging_time_h=0.96 time_to_destination_h=1.98 final_soc_percent=20.0 total_trip_time_h=3.1
```



Submitted on: 17 August 2025

Submitted by: Ryan Sequeira